Recursion - 4

Lecture-30

Raghav Garg

Finding subarrays → vo subsequence jo ki continuous fashion me ho

Find out all the subarrays of an array.

arr = {1, 2, 3, 4}

→ {}, {1}, {2}, {3}, {4}, {1,2}, {1,3}, {1,4}, {2,3}, {2,4}, {3,4},
{1,2,3}, {1,2,4}, {1,3,4}, {2,3,4}, {1,2,3,4}

1
1 2
1 2 3
1 2 3 4

2
2 3
2 3 4

3
3 4

4

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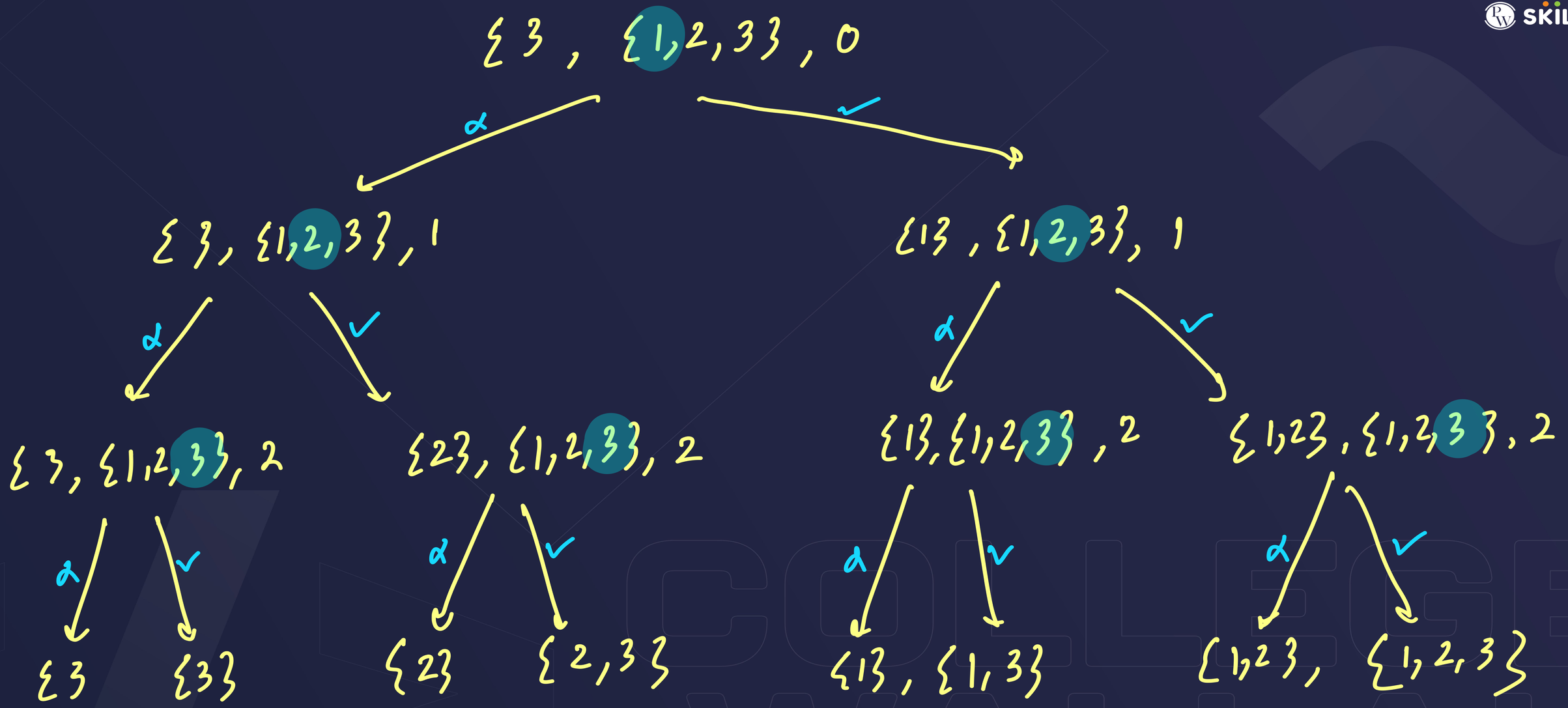
```
for(int i=0;i<n;i++){ // start
    for(int k=i;k<n;k++){
        for(int j=i;j<=k;j++){
            cout<<arr[j];
        }
        cout<<endl;
    }
}
```

k only denotes no. of rounds in loop

{1, 2, 3, 4}



Will work in unique elements only.

Palindrome using recursion

Find out whether a given string is palindrome or not using recursion

dad

mom

aba

abcdcba

racecar

Greatest Common Divisor

Calculate Greatest Common Divisor of two numbers.

hcf (24, 60) = 12

```
for(int i = 24 ; i>=2 ; i--){
    if ( 24 % i == 0 && 60 % i == 0) return i;
}
return 1;
```

→ T.C. = O(min(a, b))

hcf (a, b) <= min(a, b)

lcm (a, b) >= max(a, b)

Euclid's Division Algo (long division method)

$24 \overline{)60}\ 2$
$\quad 48$
$\quad 12$ ← HCF
$\quad 24\ 2$
$\quad 24$
$\quad 0$

$10000 \overline{)10001}\ 1$
$\quad 10000$
$\quad 1$ ← HCF
$\quad 10000\ 10000$
$\quad 10000$
$\quad 0$

$27 \overline{)45}\ 1$
$\quad 27$
$\quad 18$
$\quad 27\ 1$
$\quad 18$
$\quad 9$ ← HCF
$\quad 18\ 2$
$\quad 18$
$\quad 0$

Time Complexity of $gcd(a, b)$ is $\boxed{O(\log(a+b))}$

HCF → 27 & 45

$27 \% 45$

$a \% b = a$ if $a < b$

$\gcd(27, 45)$

$\downarrow$

$\gcd(45\%27, 27) = \gcd(18, 27)$

$\downarrow$

$\gcd(27\%18, 18) = \gcd(9, 18)$

$\downarrow$

$\gcd(18\%9, 9) = \gcd(0, 9)$

HCF

$\gcd(\underset{a}{45}, \underset{b}{27}) = \gcd(27, 45)$

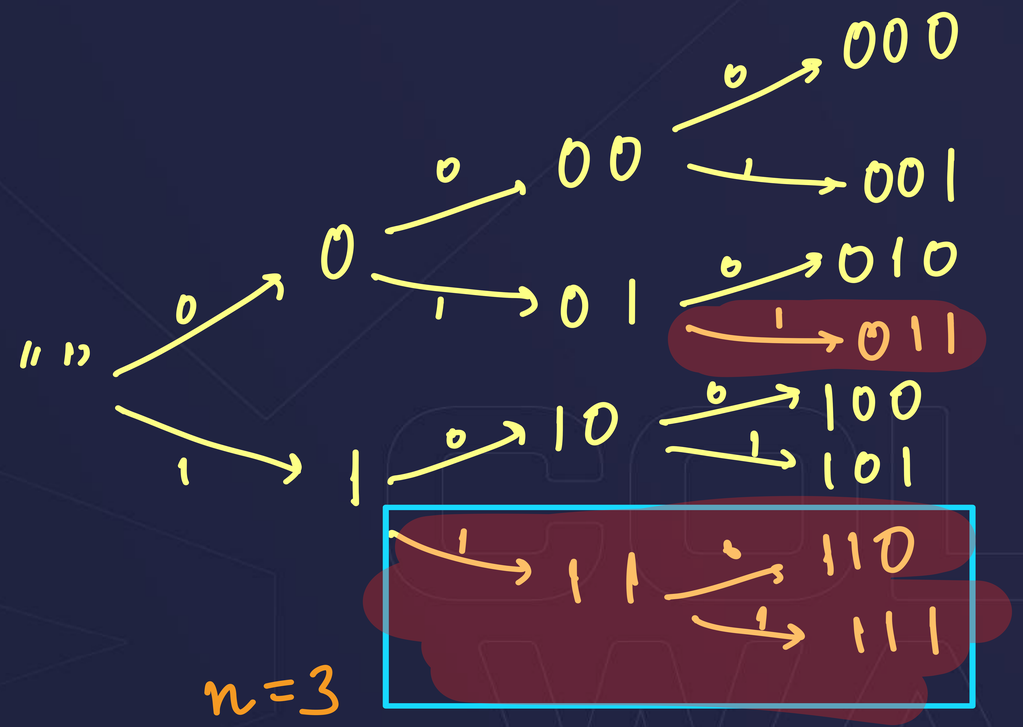
Practice

Generate all binary strings of length n without consecutive 1's

$\rightarrow 2^n$

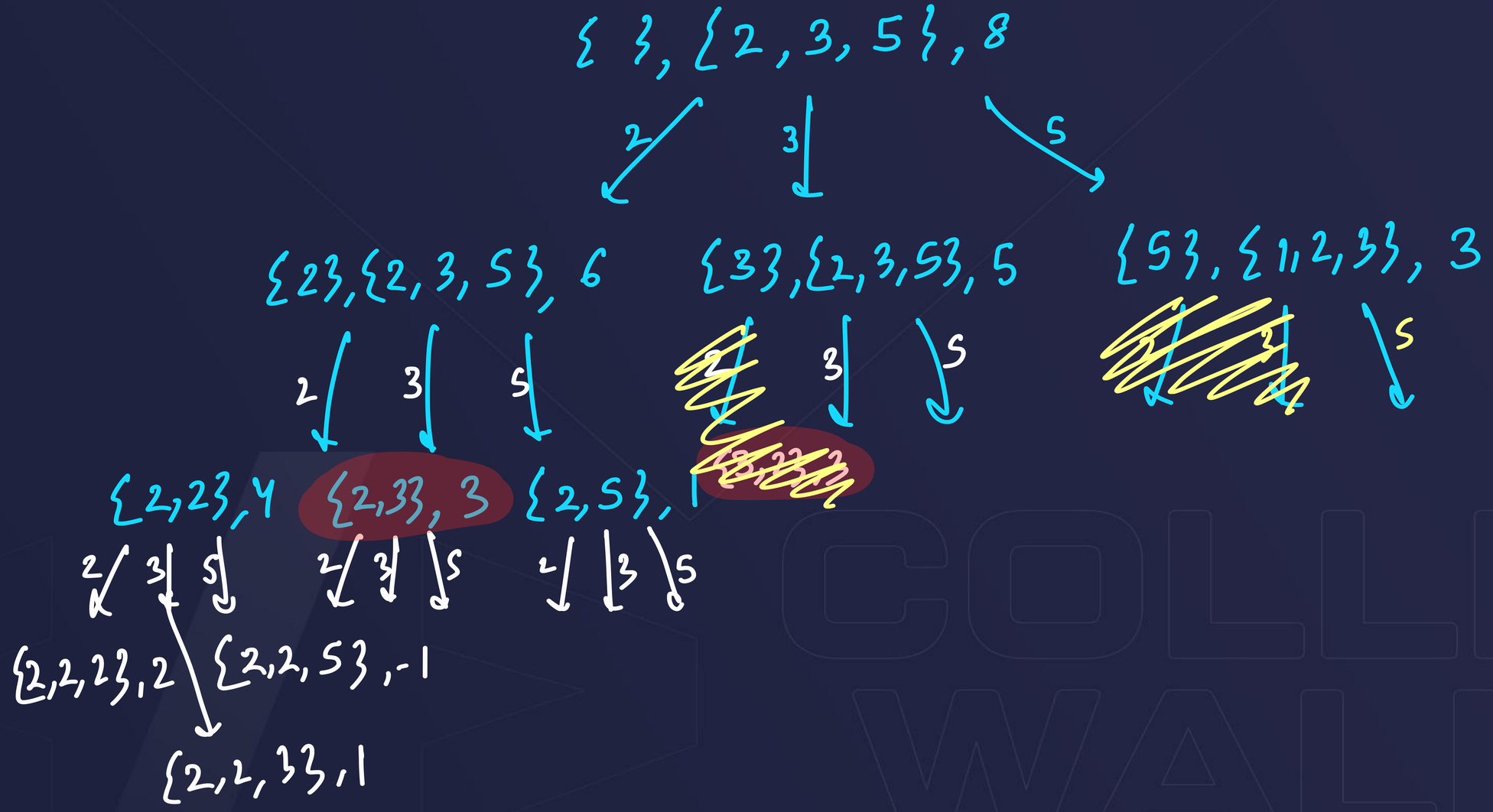
$n = 2$

$\rightarrow$ "00"
"01"
"10"
"11"



Ques: Combination Sum

[Leetcode - 39]



Ques: Generate Parentheses [Leetcode - 22]

```
Input: n = 3
Output: ["((()))","(()())","(())()","()(())","()()()"]
```

VVIMP Observations

at every instant,
no. of opening bkts >=
no. of closing bkts.

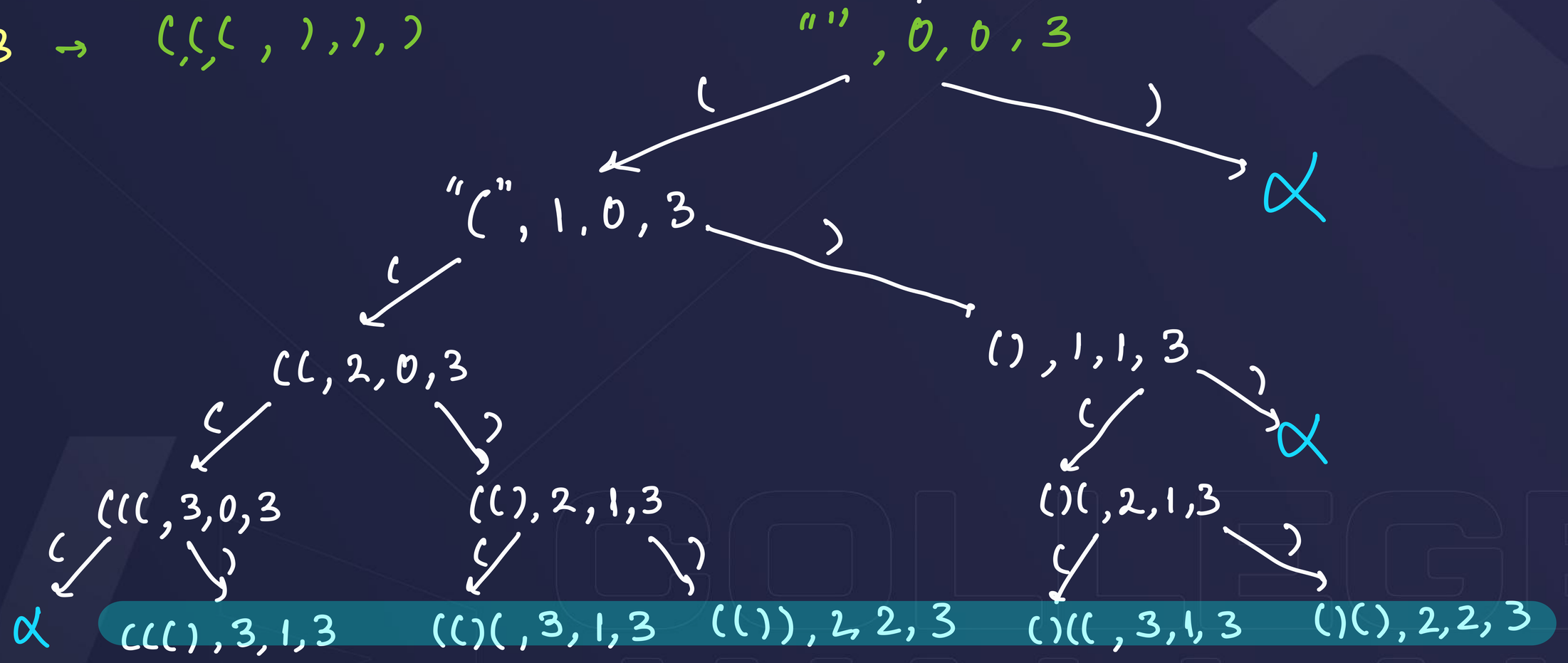
n = 1 → ( )

n = 2 → ( )( ) , (( ))

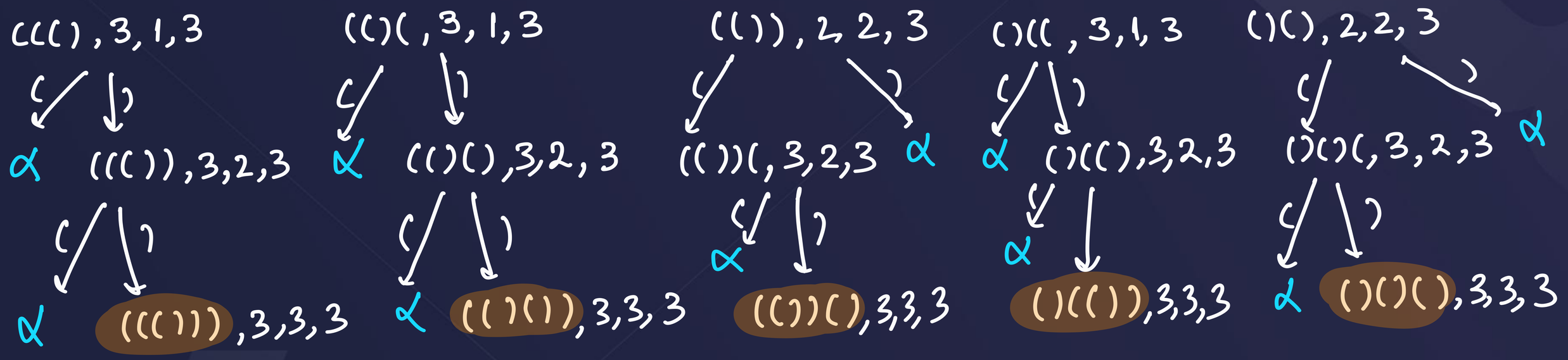
n = 3 → ( )( )( ) , (( )( )) . (( ))( ) , ((( ))) , ( )(( ))

Ques: Generate Parentheses [Leetcode - 22]

$n = 3 \rightarrow$ (((,) ,) ,)



Ques: Generate Parentheses [Leetcode - 22]




```
Input: n = 3
Output: ["((()))","(()())","(())()","()(())","()()()"]
```

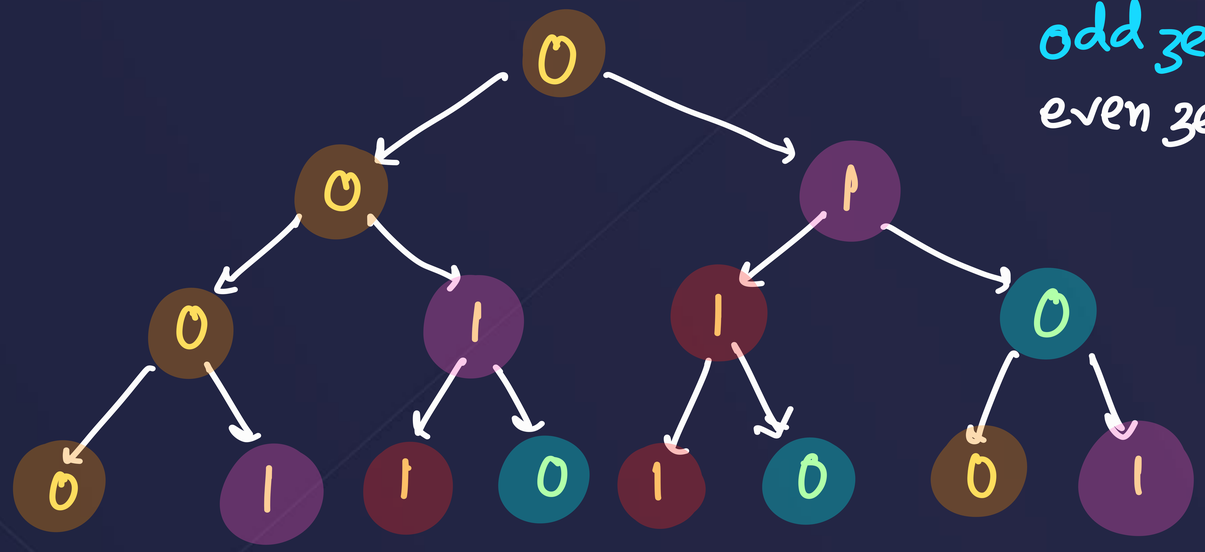
Ques: Kth Symbol in Grammar [Leetcode - 779]

1 →

2 →

3 →

4 →



odd zeroes → 0 se gen

even zeroes → 1 se gen

$\boxed{T.C. = O(n)}$

$n = 4,\ k = \cancel{6}\ 5$

$0 \to 01$

$1 \to 10$

odd ones → 1 se

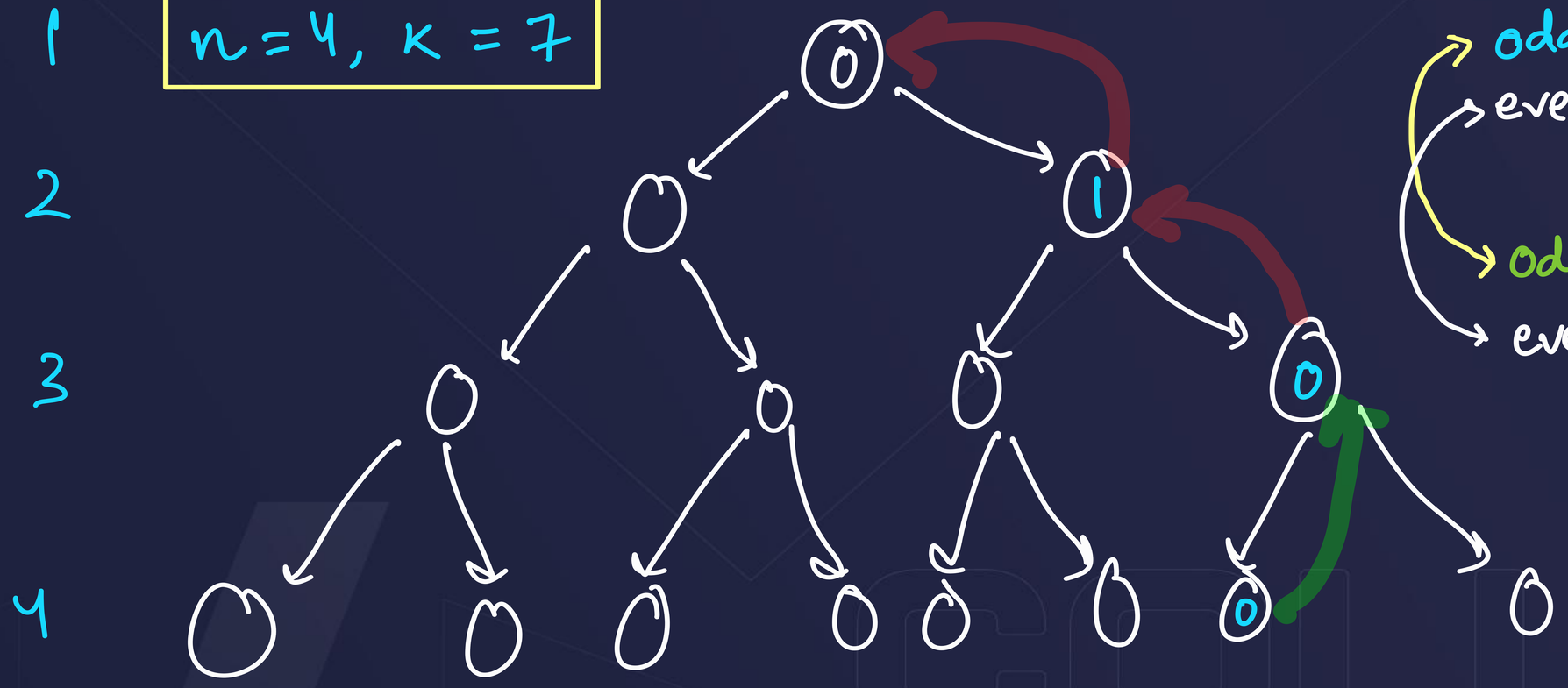
even eves → 0 se

$kth(n,k) \to if(k\%2==0) \to kth(n-1, \frac{k}{2})$

$kth(n,k) \to if(k\%2!=0) \to kth(n-1, \frac{k}{2}+1)$

Ques: Kth Symbol in Grammar [Leetcode - 779]

$n = 4,\ k = 7$



odd zeroes → 0 se gen

even zeroes → 1 se gen

odd ones → 1 se

even one → 0 se

$kth(n,k) \rightarrow if(k\%2==0) \rightarrow kth(n-1, \frac{k}{2})$

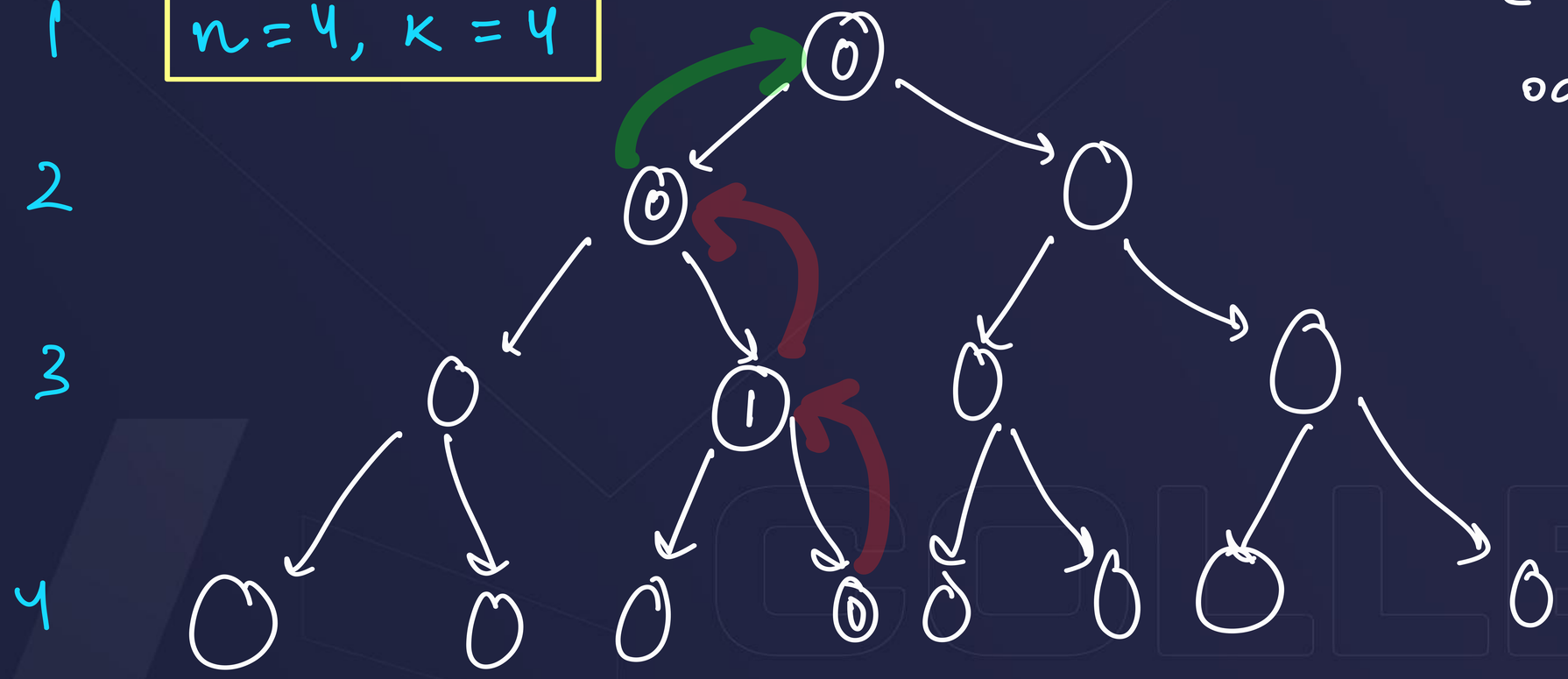
$kth(n,k) \rightarrow if(k\%2!=0) \rightarrow kth(n-1, \frac{k}{2}+1)$

Ques: Kth Symbol in Grammar [Leetcode - 779]

$n = 4, \; k = 4$

even → flip

odd → same



Ques: Count and Say

[Leetcode - 38]

cas(n) = the way you would speak cas(n-1)

cas(1) = "1";

cas(2) = one 1 → "11"

cas(3) = two 1 ⇒ "21"

cas(4) = "1211"

cas(5) = "111221"

cas(6) = "312211"

cas(7) = "13112221"

cas(8) = "1113213211"

"3322251"

two 3's, three 2's, one 5, and one 1

2 3 + 3 2 + 1 5 + 1 1

"23321511"

Ques: Permutation Sequence [Leetcode - 60]

$n = 4 \rightarrow 4! = 24$ permutations

$k = 9$

ori	n	ans	idx	k
"1234",	4,	"",	0,	9
134,	3,	2,	,	3
14,	2,	23,		

1234	3124
1243	3142
1324	3214
1342	3241
1423	3412
1432	3421
2134	4123
2143	4132
2314	4213
2341	4231
2413	4312
2431	4321

1 se start → 6

2 → $\boxed{3^{rd} \text{ string}}$

ans

2 _ _ _

1, 3, 4

134
143
314
341
413
431

→ 14

$n = 4$

$k = 18 \rightarrow fact = (n-1)! = 3! = 6$

1234	3124
1243	3142
1324	3214
1342	3241
1423	3412
1432	3421
2134	4123
2143	4132
2314	4213
2341	4231
2413	4312
2431	4321

	str	n	k	ans
→	1234	4	18	""
→	124	3	fact	3

$fact = (4-1)! = 6$

$idx = \frac{18}{6} - 1 = \boxed{2}$

```
if (k % fact != 0)  idx = k/fact ;
if (k % fact == 0)  idx = k/fact - 1
```

$n = 4$

$k = 17 \rightarrow fact = (n-1)! = 3! = 6$

1234	3124
1243	3142
1324	3214
1342	3241
1423	3412
1432	3421
2134	4123
2143	4132
2314	4213
2341	4231
2413	4312
2431	4321

	str	n	k	ans
→	1234	4	17	""
→	124	3	5	3

$fact = 6$

$idx = \frac{17}{6} = 2$

k-fact

$17 \% 6 = 5$

```
if (k % fact != 0)  idx = k/fact ;
if (k % fact == 0)  idx = k/fact - 1
```

$n = 4$

$k = 4 \rightarrow fact = (n-1)! = 3! = 6$

	str	n	k	ans
→	1234	4	4	""
→	234	3	4	1

$fact = 6$

$idx = \frac{k}{fact} = \frac{4}{6} = 0$

1234	3124
1243	3142
1324	3214
1342	3241
1423	3412
1432	3421
2134	4123
2143	4132
2314	4213
2341	4231
2413	4312
2431	4321

$n = 3$

$K = 3$

123

132

213

231

312

321

	str	n	K	ans
→	123	3	3	""
→	13	2	1	2
→	3	1	1	21

$fact = (2-1)! = 1! = 1$

$idx = \frac{K}{fact} - 1 = \frac{1}{1} - 1 = 0$

$n = 4$

$k = 2$

str	n	k	ans
1234	4	2	1111
234	3		1

$fact = 6$

$\boxed{idx} = \frac{2}{6} = 0$

```
q = 1;
if (k%fact == 0) q = fact
else (k>fact) q = k%fact
```

Next Lecture

Advanced Sorting Algorithms

THANK YOU

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